

Short Notes on UAVs and Drone Technology

PART 1

UAV vs Drone

UAV (Unmanned Aerial Vehicle) is an aircraft that operates without a human pilot on board. It may be controlled remotely or fly autonomously using pre-programmed instructions.

A **drone** is a broader term commonly used to describe UAVs along with their supporting systems such as ground control stations, communication links, sensors, and payloads.

In simple terms, every UAV is a drone, but the term drone often refers to the complete system rather than just the aircraft.

2. Types of Drones

a) **Multirotor Drone**

Multirotor drones use multiple rotors to generate lift and control movement. Common configurations include quadcopters, hexacopters, and octocopters.

Advantages:

- Vertical takeoff and landing (VTOL)
- High stability and maneuverability
- Easy to operate

Applications:

- Aerial photography
- Surveillance
- Inspection and mapping

b) **Fixed-Wing Drone**

Fixed-wing drones resemble conventional airplanes and generate lift through their wings.

Advantages:

- Longer flight endurance
- Higher speed
- Greater coverage area

Applications:

- Agricultural monitoring
- Long-range surveillance
- Environmental mapping

c) Hybrid VTOL Drone

Hybrid VTOL drones combine the characteristics of multirotor and fixed-wing aircraft. They can take off vertically and then transition to efficient forward flight.

Advantages:

- No runway required
- Long flight duration
- Efficient energy usage

Applications:

- Delivery systems
- Military reconnaissance
- Large-area surveying

3. Basic Components of a Drone

a) Flight Controller

The flight controller is the brain of the drone. It processes sensor data and controls the motors to maintain stable flight.

b) ESC (Electronic Speed Controller)

The ESC regulates the speed of the motors based on commands received from the flight controller.

c) GPS Module

The GPS module provides location, altitude, and navigation information. It enables autonomous flight, waypoint missions, and return-to-home functionality.

d) IMU (Inertial Measurement Unit)

The IMU consists of accelerometers and gyroscopes that measure acceleration, orientation, and angular velocity. It helps maintain stability during flight.

e) LiDAR / Camera Payload

LiDAR uses laser pulses to measure distances and create accurate 3D maps of the environment. Cameras capture images and videos for surveillance, inspection, and mapping applications.

4. Important Concepts

a) PID Control

PID stands for Proportional, Integral, and Derivative control. It is a feedback control mechanism used to maintain stable flight by continuously correcting errors between the desired and actual state of the drone.

b) MAVLink

MAVLink (Micro Air Vehicle Link) is a lightweight communication protocol used for communication between drones, ground control stations, onboard computers, and companion devices. It enables telemetry, command transmission, and mission control.

c) Autopilot Firmware

Autopilot firmware is the software running on the flight controller that manages flight operations, navigation, stabilization, and mission execution.

Examples include:

- ArduPilot
- PX4

The firmware interprets sensor data and controls the drone autonomously or under operator commands.

PART 3:

Thrust

Thrust is the upward force produced by the drone's propellers. It allows the drone to lift off the ground and stay airborne.

Lift

Lift is the force that opposes gravity and keeps the drone in the air. In quadcopters, lift is generated by the rotating propellers.

Drag

Drag is the air resistance that opposes the movement of the drone. Higher drag reduces speed and efficiency.

Pitch

Pitch refers to the forward or backward tilt of the drone. Increasing the speed of the rear motors causes the drone to tilt forward.

Roll

Roll refers to the left or right tilt of the drone. It is controlled by varying the speed of motors on opposite sides.

Yaw

Yaw is the rotation of the drone around its vertical axis. It allows the drone to change its heading direction.

Calculation

Given:

- Drone Weight = 1.5 kg
- Number of Motors = 4

Step 1: Calculate Total Thrust Required

For takeoff:

$$Total\ Thrust > Weight$$

Minimum thrust required:

$$\textit{Total Thrust} = 1.5 \text{ kg}$$

Step 2: Thrust Required Per Motor

$$\begin{aligned}\textit{Thrust per motor} &= \frac{1.5}{4} \\ &= 0.375 \text{ kg}\end{aligned}$$

Industry Practice

A safety factor of 2 is generally used.

$$\begin{aligned}\textit{Required Total Thrust} &= 3 \text{ kg} \\ \textit{Required Thrust per Motor} &= \frac{3}{4} \\ &= 0.75 \text{ kg}\end{aligned}$$

Conclusion

For a quadcopter weighing 1.5 kg, each motor should provide a minimum thrust of 0.375 kg for takeoff. In practical applications, a safety factor is applied, increasing the recommended thrust to approximately 0.75 kg per motor for stable and efficient flight.

PART 4

Explanation for Report

Objective

The objective of this simulation was to understand how thrust affects the altitude of a quadcopter. A simple physics model was implemented in Python to simulate the vertical motion of a UAV under the influence of thrust and gravity.

Working Principle

The simulation assumes that the drone moves only in the vertical direction. Two forces act on the drone:

1. **Thrust** generated by the motors (upward force)
2. **Weight** of the drone due to gravity (downward force)

The net force acting on the drone is calculated as:

$$F_{\text{net}} = F_{\text{thrust}} - mg$$

where:

- F_{thrust} = Total thrust produced by motors
- m = Mass of drone
- g = Gravitational acceleration

Using Newton's Second Law,

$$a = F_{\text{net}}/m$$

the acceleration is calculated. This acceleration is then used to update the velocity and altitude of the drone over time.



```
import numpy as np
import matplotlib.pyplot as plt

# Constants
mass = 1.5          # kg
gravity = 9.81      # m/s^2
thrust = 20         # Newton (constant thrust)
dt = 0.1

time = np.arange(0, 10, dt)

velocity = 0
height = 0

heights = []

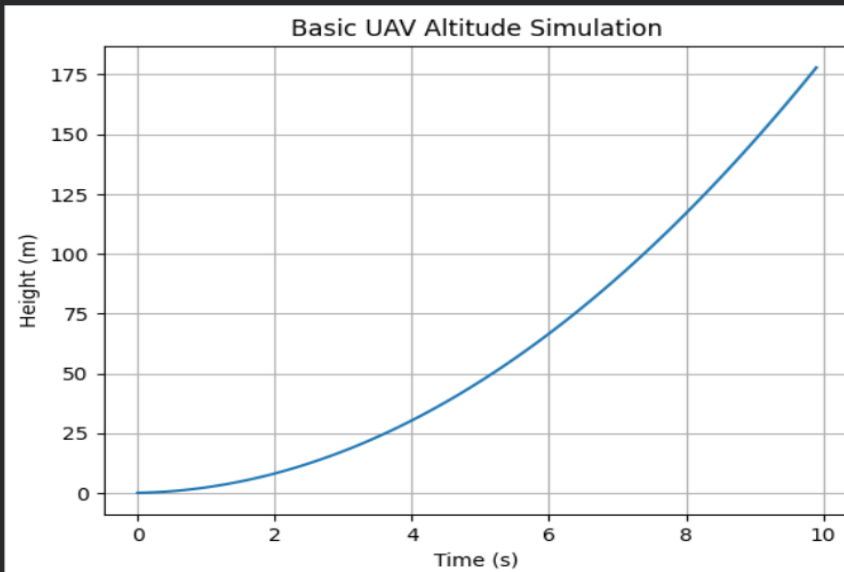
for t in time:
    net_force = thrust - (mass * gravity)
    acceleration = net_force / mass

    velocity += acceleration * dt
    height += velocity * dt

    heights.append(height)

plt.plot(time, heights)
plt.xlabel("Time (s)")
plt.ylabel("Height (m)")
plt.title("Basic UAV Altitude Simulation")
plt.grid(True)
plt.show()
```

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Part 5: Industry Trends in UAV Technology

Introduction

Drone technology is growing rapidly and is being used in many fields beyond photography. Advances in artificial intelligence, automation, and communication systems are making UAVs smarter and more capable. Some important trends are discussed below.

1. AI-Enabled UAVs

Artificial Intelligence helps drones make decisions on their own. AI-enabled UAVs can identify objects, avoid obstacles, and navigate without constant human control. They are widely used in surveillance, inspections, and search-and-rescue operations.

2. Swarm Drone Systems

Drone swarms consist of multiple UAVs working together to complete a mission. These systems can cover larger areas and continue operating even if one drone fails. Swarm technology is being explored for defense, disaster response, and environmental monitoring.

3. Drone Delivery Logistics

Many companies are testing drones for package and medical supply delivery. Drone delivery can reduce travel time, improve efficiency, and reach remote locations where traditional transportation may be difficult.

4. Agricultural Spraying Drones

Drones are increasingly used in agriculture to monitor crops and spray fertilizers or pesticides. They help farmers cover large areas quickly while reducing chemical wastage and labor costs.

5. Defense Surveillance Drones

Defense forces use UAVs for reconnaissance, border monitoring, and intelligence gathering. These drones can operate in difficult environments and provide real-time information without putting personnel at risk.

Conclusion

UAV technology is expanding into various industries, including agriculture, logistics, and defense. Developments in AI, drone swarms, and autonomous operations are making drones more efficient and capable. As technology continues to improve, UAVs are expected to play an even bigger role in both civilian and military applications.